

External Evaluation Metrics

Mutual Information

It measures how much information the clusters share with the actual class labels (true labels). A high Mutual Information indicates that the clusters tell us a lot about the true labels.

$$I(C; T) = \sum_{i=1}^K \sum_{j=1}^C P(C_i, T_j) \cdot \log \left(\frac{P(C_i, T_j)}{P(C_i) \cdot P(T_j)} \right)$$

Default base: 2
(cons in bits / shannons)

If base: 10
(cons in dits / hartley)

C : random variable representing the cluster assignments

T : random variable representing the true label classes

$P(C_i, T_j) = P(C=i \text{ and } T=j)$: joint probability of data points being in both i^{th} cluster and j^{th} class

$P(C_i) = P(C=i)$: marginal probability of data points being in i^{th} cluster

$P(T_j) = P(T=j)$: marginal probability of data points being in j^{th} true class.

eg.

	class A	class B	class C	rowSum
cluster 1	5	1	0	6
cluster 2	1	3	1	5
cluster 3	0	2	3	5
column sum	6	6	4	16

Calculate the mutual information for the clustering dataset given.

Sol.

For cluster 1, class A: $P(C=1, T=A) \cdot \log \frac{P(C=1, T=A)}{P(C=1) \cdot P(T=A)}$

$$P(C=1, T=A) = \frac{5}{16}, \quad P(C=1) = \frac{6}{16}, \quad P(T=A) = \frac{6}{16}$$

$$MI = \frac{5}{16} \times \log_2 \frac{5/16}{6/16 \times 6/16} = \frac{5}{16} \times \log_2 \left(\frac{5 \times 16}{6 \times 6} \right) \approx 0.36 \text{ bits}$$

Similarly we can calculate for every cluster, class pair and add them up to get overall mutual information.

Normalized Mutual Information

Normalizes the MI value between $[0, 1]$ making it easier to interpret.

A value of 1 indicates perfect clustering, 0 means it is no better than random clustering.

$$NMI = \frac{2 \times I(C, T)}{H(C) + H(T)}$$

$I(C, T)$: mutual information b/w the clusters and true labels

$H(C)$: Entropy of clusters

$H(T)$: Entropy of true class labels

$$H(C) = - \sum_{i=1}^K P(C_i) \cdot \log P(C_i) \quad , \quad H(T) = - \sum_{i=1}^C P(T_i) \cdot \log P(T_i)$$

eg. 7

	class A	class B	class C	rowSum
cluster 1	5	1	0	6
cluster 2	1	3	1	5
cluster 3	0	2	3	5
column sum	6	6	4	16

Calculate the NMI for the clustering dataset given.

Sol. $H(C) = - \sum_{i=1}^K P(C_i) \cdot \log P(C_i)$

$$= - \left[\frac{6}{16} \times \log_2 \frac{6}{16} + \frac{5}{16} \times \log_2 \frac{5}{16} + \frac{5}{16} \times \log_2 \frac{5}{16} \right]$$

$$\approx 1.58$$

$$H(T) = - \sum_{i=1}^C P(T_i) \cdot \log P(T_i)$$

$$= - \left[\frac{6}{16} \times \log_2 \frac{6}{16} + \frac{6}{16} \times \log_2 \frac{6}{16} + \frac{4}{16} \times \log_2 \frac{4}{16} \right]$$

$$\approx 1.56$$

$$NMI = \frac{2 \times I(C, T)}{1.58 \times 1.56} \approx \boxed{0.373} \quad (H/W)$$